

In this manuscript, we investigate Finslerian exponentially harmonic functions. Analogous to the Riemannian case, a function u defined on a Finsler metric measure space (M, F, μ) is said to be exponentially harmonic if it satisfies

$$\tilde{\Delta}_\mu u := \operatorname{div}_\mu(V(u)Du) = 0, \quad (1.1)$$

where $V(u) = \exp(\frac{1}{2}F^{*2}(Du))$ represents the energy density of u . Such a function arises as a critical point of the exponential energy functional (see Theorem 3.1). Finsler metric measure spaces encompass a richer set of geometric tensors compared to Riemannian manifolds (cf. [11]). It is important to emphasize that (1.1) is not merely a quasilinear elliptic equation, as the underlying space is an anisotropic and asymmetric manifold. Within this framework, we establish the following theorem.

Theorem 1.1. *Let (M, F, μ) be a forward complete n -dimensional Finsler metric measure space with finite misalignment α . Assume that the mixed weighted Ricci curvature ${}^m\operatorname{Ric}^\infty$ of M is nonnegative, and that the S -curvature as well as the non-Riemannian curvatures U , $\mathcal{T}$ and $\operatorname{div}C = FC^i_{jk|i}dx^j \otimes dx^k$ satisfy the norm bound $F^{-1}|S| + F(U) + F(\mathcal{T}) + \|\operatorname{div}C\|_{HS(V)} \leq K_0$, for any $(x, V) \in SM$. Then, any bounded exponentially harmonic function u on M is constant.*

2 Related concepts and notations of the Finsler metric measure spaces

A Finsler metric measure space is a triple (M, F, μ) , where M is a differentiable manifold equipped with a Finsler metric F and a Borel measure μ . The *Cartan tensor* is defined by

$$C(X, Y, Z) := C_{ijk}X^iY^jZ^k = \frac{1}{4} \frac{\partial^3 F^2(x, y)}{\partial y^i \partial y^j \partial y^k} X^i Y^j Z^k,$$

for any local vector fields X, Y, Z . We always adopt the Chern connection, which is uniquely determined by

$$\begin{aligned} \nabla_X Y - \nabla_Y X &= [X, Y]; \\ Z(g_y(X, Y)) - g_y(\nabla_Z X, Y) - g_y(X, \nabla_Z Y) &= 2C_y(\nabla_Z y, X, Y), \end{aligned}$$

for any $X, Y, Z \in TM \setminus \{0\}$, where C_y is the Cartan tensor. The coefficients of the Chern connection are locally expressed as $\Gamma^i_{jk}(x, y)$ in natural coordinates. These coefficients induce the *spray coefficients* as $G^i = \frac{1}{2}\Gamma^i_{jk}y^jy^k$. The spray is given by

$$G = y^i \frac{\delta}{\delta x^i} = y^i \frac{\partial}{\partial x^i} - 2G^i \frac{\partial}{\partial y^i}, \quad (2.1)$$

where $\frac{\delta}{\delta x^i} = \frac{\partial}{\partial x^i} - N^j_i \frac{\partial}{\partial y^j}$, and the nonlinear connection coefficients N^i_j are locally derived from the spray coefficients as $N^i_j = \frac{\partial G^i}{\partial y^j}$. By convention, the horizontal Chern derivative is denoted by “ $|$ ” and the vertical Chern derivative by “ $,$ ”. Let $\hat{D}$ denote the Levi-Civita connection of the induced Riemannian metric $\hat{g} = g_Y$, and let $\{e_i\}$